Assignment 3

Q1. Implementation and Analysis of Bubble Sort in C: Best, Worst, and Average Case Performance Using GNUplot.

Implement Bubble Sort in C. Measure its execution time across best, worst, and average case input arrangements for varying input sizes n. Plot the resulting growth curves using GNUplot to analyze its empirical time complexity.

Q2. Implementation and Comparative Analysis of Standard Bubble Sort and Optimized Bubble Sort in C (Best Case) Using GNUplot.

Implement both standard Bubble Sort and an optimized version (incorporating an early-exit flag for sorted arrays) in C. Measure and compare their execution times on best-case (pre sorted) input data across varying input sizes n. Use GNUplot to graph the performance comparison and demonstrate O(n) vs. O(n2) best-case behavior.

Q3. Implementation and Analysis of Insertion Sort in C: Best, Worst, and Average Case Performance Using GNUplot.

Implement Insertion Sort in C. Measure and record execution times for best, worst, and average-case input configurations across varying input sizes n. Plot the execution time curves using GNUplot to analyze its theoretical vs. empirical performance